

Existence of a Model Companion for Groups of Exponent 3

Yawara Ishida*, Ryosuke Mizuno[†] and Kota Takeuchi[‡]

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Abstract

In this article, we prove that the theory T_3 of groups of exponent 3 has a model companion. Previous work established the existence of model companions for theories of groups of fixed finite exponent and nilpotency class at most 2 (Saracino–Wood), and for theories of groups of prime exponent p and nilpotency class at most $c < p$ (Maier). These results do not cover T_3 , since groups of exponent 3 may have nilpotency class 3. Our theorem verifies the $n = 3$ case of a conjecture proposed by the third author that, for each integer $n > 1$, the theory T_n of groups of exponent n has a model companion if and only if every finitely generated group of exponent n is finite, or equivalently, if and only if the Burnside problem has a positive solution for exponent n .

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Use of generative AI. An initial outline of the proof of the main theorem was generated by ChatGPT (OpenAI, GPT-5.6 Sol). The authors subsequently developed and reconstructed the argument and checked all proofs in detail. The authors take full responsibility for the content of this paper.

1 Introduction

The question of whether a given theory has a model companion has been studied for many classes of groups. In the abelian case, many standard theories have model companions. This is known for the theory of all abelian groups, the theory of torsion-free abelian groups, and the theory of abelian groups of any fixed finite exponent. More generally, Eklof proved that every inductive theory of abelian groups with the joint embedding property has a model companion [Ekl1972].

The nonabelian case is less uniform. The theory of all groups has no model companion [CK1990, Example 3.5.16]. The same is true for the theory of torsion-free groups; see, for example, [Tak2022]. Saracino obtained non-existence results for the theories of solvable groups of fixed derived length at least 2 [Sar1974]. He also proved that, for every $c \geq 2$, neither

*A.I. Systems Research Institute Co., Ltd.

[†]A.I. Systems Research Institute Co., Ltd.

[‡]Institute of Mathematics, University of Tsukuba. Partially supported by JSPS KAKENHI Grant Number 25K00213. kota@math.tsukuba.ac.jp

the theory of nilpotent groups of class at most c nor its torsion-free version has a model companion [Sar1976]. Thus, standard conditions such as solvability or a fixed bound on nilpotency class are not enough to ensure the existence of a model companion.

Finite exponent changes the answer for some nilpotent classes. Saracino and Wood proved that the theory of groups of fixed finite exponent and nilpotency class at most 2 has a model companion [SW1979]. For prime exponent p and nilpotency class at most $c < p$, Maier proved that the theory has an $\aleph_0$ -categorical model companion [Mai1989]; see also [dEMRS2025, Corollary 4.42].

There are also positive results of a different type. For example, recent work gives model-complete pure group theories for some semisimple algebraic groups and for Heisenberg groups over model-complete fields [HKTY2023, FK2025]. These results show that noncommutativity itself is not an obstruction. However, they concern complete theories of particular groups. They do not give a criterion for broad classes of groups defined by simple group-theoretic axioms. In contrast with the abelian case, no general result is known that describes the boundary between existence and non-existence for such nonabelian classes.

One way to study this boundary is to consider universal theories, and in particular varieties of groups. The results above suggest that finite exponent is an important condition to examine. For an integer $n > 1$, let $\mathcal{V}_n$ be the variety of groups satisfying $x^n = 1$, and let T_n be its theory in the usual language of groups. The axioms defining T_n impose a fixed exponent identity but no separate bound on nilpotency class or derived length. It is therefore a natural test case for the model-companion problem for nonabelian group varieties.

We prove the following.

Main Theorem. *The theory T_3 has a model companion.*

The case $n = 2$ is abelian. Thus, $n = 3$ is the first nonabelian case in this family. Groups of exponent 3 are nilpotent of class at most 3, and this bound is attained [LW1933]. Hence our result concerns the whole variety $\mathcal{V}_3$, including its groups of nilpotency class 3. It is not covered by the class-2 result of Saracino and Wood or by the results for prime exponent p and class $c < p$.

The family $(T_n)_{n>1}$ is also related to the Burnside problem. The possible connection between the model-companion problem for groups of fixed exponent and the Burnside problem was already noted by the third author in [Tak2022]. The following conjecture, proposed by the third author, gives a precise formulation of this connection.

Conjecture 1.1. *For every integer $n > 1$, the theory T_n has a model companion if and only if the Burnside problem has a positive solution for exponent n .*

The condition on the right says that $\mathcal{V}_n$ is locally finite, or equivalently, every finitely generated group satisfying $x^n = 1$ is finite. The conjecture holds for $n = 2$ by the theory of vector spaces over $\mathbb{F}_2$. The Burnside problem also has a positive solution for exponent 3, and the main theorem proves the other side of the conjecture in this case. We have also obtained a proof that T_p has no model companion for every sufficiently large prime p . Since the Burnside problem has a negative solution for every sufficiently large prime p by the Novikov–Adian theorem, and in fact for every odd exponent at least 665 [NA1968, Adi1979], this proves the conjecture for all sufficiently large primes. The proof will appear elsewhere.

We briefly explain the main idea of the proof. The variety $\mathcal{V}_3$ is locally finite, but it does not have the amalgamation property. We therefore study finite witnesses to failure of amalgamation.

Let M be an existentially closed model of T_3 , let $A \leq M$, and let $A \leq B \in \mathcal{V}_3$ be finite. If M and B have no amalgam over A , we show that the same failure already occurs between B and a finite subgroup D of M containing A , where the number of generators of D is bounded in terms of the number of generators of B . In other words, the failure of amalgamation in the coproduct $M *_{\mathcal{V}_3} B$ can be reproduced in a coproduct involving only boundedly many generators. By Fact 2.6, this uniform bound gives the model companion.

For this purpose, we prove a finite strict-envelope theorem. Every finite subgroup C of an existentially closed model M is contained in a finite subgroup $D \leq M$ that is strict with respect to the lower central series, with a quadratic bound on the number of generators of D . This strictness implies that the natural map

$$D *_{\mathcal{V}_3} B \longrightarrow M *_{\mathcal{V}_3} B$$

is injective. Hence a witness to the failure of amalgamation that is found in $M *_{\mathcal{V}_3} B$ and involves only elements of D and B is already a witness in $D *_{\mathcal{V}_3} B$. We also prove a structure theorem for existentially closed models of T_3 : their lower and upper central series coincide in reverse order, and every element of the second and third terms of the lower central series is a commutator or a triple commutator, respectively. The main tool for these results is the associated graded Lie algebra. We compute the graded Lie algebras of coproducts in $\mathcal{V}_3$ and use these computations to construct the required extensions.

Section 2 contains the preliminaries. Section 3 proves the main theorem from the bounded-witness and strict-envelope results. Section 4 proves the structural results used in the argument. Section 5 gives examples showing that lower-central strictness and existential closedness are needed.

We have also fully formalized all the results in this paper in Lean 4. The formalization depends only on the three standard axioms (`propext`, `Classical.choice`, and `Quot.sound`), with no `sorryAx` or project-specific axioms. The formalization will be released on GitHub soon.

2 Preliminaries

Throughout, groups are regarded as structures in the language $\mathcal{L}_{\text{grp}} = \{\cdot, {}^{-1}, 1\}$. Let T_{grp} denote the theory of groups. For an integer $n > 1$, set

$$T_n := T_{\text{grp}} \cup \{\forall x (x^n = 1)\}.$$

Thus, by a group of exponent n , we mean a group whose exponent divides n .

We fix the following notation for group elements:

Notation 2.1. 1. $[a, b] = aba^{-1}b^{-1}$

2. $[a_1, \dots, a_k] = [[a_1, \dots, a_{k-1}], a_k]$

3. $a^b = b^{-1}ab$

4. $\mathcal{V}_3$ is the variety of groups satisfying $x^3 = 1$; equivalently, $\mathcal{V}_3 = \{M \mid M \models T_3\}$.

5. $G * H$ is the free product of groups G and H .

6. For a subset S of a group G , $\langle\langle S \rangle\rangle_G$ denotes the normal closure of S in G . We omit the subscript when the ambient group is clear.

7. $d(G)$ is the minimum number of generators of G .

8. $G *_{\mathcal{V}_3} H$ is the coproduct of G and H in the variety $\mathcal{V}_3$. Equivalently,

$$G *_{\mathcal{V}_3} H = (G * H) / \langle\langle (G * H)^3 \rangle\rangle_{G * H}.$$

9. For a set X , $F_{\mathcal{V}_3}(X)$ is the free group in the variety $\mathcal{V}_3$ generated by X . Equivalently, if $F(X)$ is the free group on X , then

$$F_{\mathcal{V}_3}(X) = F(X) / \langle\langle F(X)^3 \rangle\rangle_{F(X)}.$$

If $|X| = r \in \mathbb{N}$, then $F_{\mathcal{V}_3}(X)$ is called the Burnside group $B(r, 3)$.

2.1 Model-theoretic background

Let T and T' be L -theories.

Definition 2.2. 1. T_0 and T_1 are called companions if for $i = 0, 1$, for every $M_i \models T_i$ there is $M_{1-i} \models T_{1-i}$ such that $M_i \subset M_{1-i}$.

2. T is said to be model complete if for every formula $\theta(\bar{x})$ there is an existential formula $\psi(\bar{x})$ such that $T \models \forall \bar{x}(\theta(\bar{x}) \leftrightarrow \psi(\bar{x}))$.

3. T^* is called a model companion of T if T^* is a companion of T and is model complete.

4. $M \models T$ is an existentially closed model (e.c. model) of T if for any $\bar{a} \in M$ and any extension $M \subset N \models T$, $N \models \exists \bar{x}\varphi(\bar{x}, \bar{a})$ implies $M \models \exists \bar{x}\varphi(\bar{x}, \bar{a})$ where $\varphi(\bar{x}, \bar{y})$ is quantifier-free.

5. T is a Π_2 -theory if there is a set T' of $\forall\exists$ -sentences such that $T \equiv T'$.

6. Let $M \models T$ and let $A \subset M$ be a finite subset. Set

$$\text{Diag}_A(\bar{x}) = \{\theta(\bar{x}) \mid M \models \theta(A), \theta(\bar{x}) \text{ is quantifier-free}\},$$

where $|\bar{x}| = |A|$. If L is finite and T is locally finite, then $\text{Diag}_A(\bar{x})$ is finite modulo T .

Fact 2.3. Let T be a Π_2 -theory. Then for an L -theory T^* , T^* is a model companion of T if and only if

$$\{M \mid M \models T^*\} = \{M \models T \mid M \text{ is an e.c. model of } T\}.$$

Definition 2.4. Let T be an L -theory. We say T is locally finite if, for every $M \models T$ and every finite subset $A \subset M$, the substructure $\langle A \rangle \subset M$ generated by A is finite.

The next fact follows from a standard compactness argument.

Fact 2.5. Let L be finite and let T be a locally finite L -theory. Then for any $n \in \mathbb{N}$, there is $m \in \mathbb{N}$ such that if $A \subset M \models T$ is a subset with $|A| \leq n$, then $|\langle A \rangle| \leq m$.

Fact 2.6. Let L be finite and let T be a locally finite L -theory. Suppose that T is a Π_2 -theory. Then the following are equivalent.

1. T has a model companion T^* .

2. For any finite substructures $A \subset B$ of a model $M_0 \models T$, there is $n_{A,B} \in \mathbb{N}$ satisfying the following: Let M be an e.c. model of T and let $A \subset M$. If M and B do not have an amalgam over A in T , then there is $A \subset C \subset M$ such that

- (a) C is generated by at most $n_{A,B}$ elements as an L -substructure,
- (b) C and B do not have an amalgam over A in T .

Proof. For given finite $A \subset B$, let $\theta_{B,A}(\bar{x}, A)$ be a quantifier-free $L(A)$ -formula expressing $\text{Diag}_{AB}(\bar{x}, \bar{y})$; in other words, $\theta_{B,A}(B', A')$ holds if and only if $A \cong A'$ and $B \cong_A B'$ after identifying A' with A . Since L is finite, we can find such a formula $\theta_{B,A}(\bar{x}, A)$.

(1 $\Rightarrow$ 2) Since T^* is a model companion, $T^* \models \forall \bar{y}(\neg \exists \bar{x} \theta_{B,A}(\bar{x}, \bar{y}) \leftrightarrow \exists \bar{z} \psi(\bar{z}, \bar{y}))$ for some quantifier-free L -formula $\psi(\bar{z}, \bar{y})$. Let $n_{A,B} = |\bar{z}| + |A|$.

Suppose M is an e.c. model of T and $A \subset M$. If M and B do not have an amalgam over A in T , then $M \models \neg \exists \bar{x} \theta_{B,A}(\bar{x}, A)$. Since $M \models T^*$, $M \models \exists \bar{z} \psi(\bar{z}, A)$. Let $\bar{c} \in M$ be a realization of $\psi(\bar{z}, A)$ in M and let $C = \langle \bar{c}, A \rangle_M \subset M$. Suppose that C and B have an amalgam $N_0 \models T$ over A . We can extend $N_0 \subset N \models T^*$. Then $N \models \psi(\bar{c}, A)$, hence $N \models \neg \exists \bar{x} \theta_{B,A}(\bar{x}, A)$. However, N contains a copy of B over A , which implies $\exists \bar{x} \theta_{B,A}(\bar{x}, A)$.

(2 $\Rightarrow$ 1) For each finite $A \subset B$, let $K(A, B)$ be the set of all (isomorphism types of) $C \supset A$ over A such that

- 1. C is a substructure of some $N \models T$ and C is generated by at most $n_{A,B}$ elements,
- 2. C and B do not have an amalgam in T over A .

Because T is locally finite, the size $|C|$ is uniformly bounded by some $m_{A,B} \in \mathbb{N}$. Hence $K(A, B)$ is finite since L is finite. Let $\theta_A(\bar{y})$ be a quantifier-free formula expressing $\text{Diag}_A(\bar{y})$, and let

$$T^* = T \cup \left\{ \forall \bar{y} \left[\theta_A(\bar{y}) \wedge \left(\bigwedge_{C \in K(A,B)} \neg \exists \bar{z} \theta_{C,A}(\bar{z}, \bar{y}) \right) \rightarrow \exists \bar{x} \theta_{B,A}(\bar{x}, \bar{y}) \right] \mid A \subset B \text{ are finite} \right\}.$$

If $M \models T$ is an e.c. model, then clearly $M \models T^*$, because $A \subset M$ and $A \subset B$ have an amalgam in T over A if and only if $M \models \exists \bar{x} \theta_{B,A}(\bar{x}, A)$. Conversely, suppose $M \models T^*$ and let $\bar{a} \subset M \subset N \models T$, $\varphi(\bar{x}, \bar{a})$ a quantifier-free formula, and $N \models \exists \bar{x} \varphi(\bar{x}, \bar{a})$. Then there is $\bar{b} \in N$ such that $N \models \varphi(\bar{b}, \bar{a})$, so let $B = \langle \bar{b}, \bar{a} \rangle_N$ and $A = \langle \bar{a} \rangle_M$. This means N is an amalgam of M and B over A . For every $C \in K(A, B)$, the definition of $K(A, B)$ gives $N \models \neg \exists \bar{z} \theta_{C,A}(\bar{z}, A)$. Because $M \subset N$ and $\theta_{C,A}$ is quantifier-free, $M \models \neg \exists \bar{z} \theta_{C,A}(\bar{z}, A)$. Hence we can find $B' \subset M \models \theta_{B,A}(B', A)$ and $\bar{b}' \in B'$ such that $M \models \varphi(\bar{b}', \bar{a})$. $\square$

2.2 Graded Lie algebras and block decomposition

Definition 2.7. A graded Lie ring L is an abelian group $(L, +)$ with a decomposition $L = \bigoplus_{i \geq 1} L_i$ and a Lie bracket $[-, -] : L \times L \rightarrow L$ satisfying the following:

- 1. $[L_i, L_j] \subset L_{i+j}$
- 2. $[-, -]$ is biadditive: $[a + b, c] = [a, c] + [b, c]$ and $[a, b + c] = [a, b] + [a, c]$
- 3. $[a, a] = 0$

4. Jacobi identity: $[[a, b], c] + [[b, c], a] + [[c, a], b] = 0$

If L is a vector space over a field k and $[-, -]$ is k -bilinear, then L is called a graded Lie algebra over k .

Remark 2.8. 1. $[a, b] = -[b, a]$ for $a, b \in L$.

2. L is generated by L_1 if and only if $[L_i, L_1] = L_{i+1}$ for every $i \geq 1$.

Example 2.9 (Grassmann algebra). Let V be a vector space over $\mathbb{F}_3$, and set

$$\Lambda^{[1,3]}V := \Lambda^1V \oplus \Lambda^2V \oplus \Lambda^3V.$$

For homogeneous elements $u \in \Lambda^iV$ and $v \in \Lambda^jV$, define

$$[u, v] := \begin{cases} u \wedge v, & (i, j) = (1, 1) \text{ or } (2, 1), \\ -u \wedge v, & (i, j) = (1, 2), \\ 0, & i + j \geq 4. \end{cases}$$

Extending this bracket bilinearly makes $\Lambda^{[1,3]}V$ a graded Lie algebra over $\mathbb{F}_3$.

Definition 2.10. Let V be a vector space and fix a decomposition $V = \bigoplus_{\lambda} V_{\lambda}$. Each V_{λ} is called a block or λ -block. A subspace $W \subset V$ is called block-homogeneous (with respect to the decomposition) if $W = \bigoplus_{\lambda} (W \cap V_{\lambda})$.

Proposition 2.11. Let $V = \bigoplus_{\lambda} V_{\lambda}$ be a decomposition of a vector space and let $W \subset V$ be a block-homogeneous subspace of V . Then $V/W \cong \bigoplus_{\lambda} V_{\lambda}/(W \cap V_{\lambda})$ with a standard isomorphism $\bar{v} \mapsto \bigoplus_{\lambda} \bar{v}_{\lambda}$ where $v = \bigoplus_{\lambda} v_{\lambda} \in V$.

Proof. Straightforward. □

2.3 Group-theoretic background

Let G be a group. For subsets $A, B \subset G$, let $[A, B]$ denote the subgroup generated by $\{[a, b] : a \in A, b \in B\}$.

Definition 2.12 (Lower and upper central series). 1. The lower central series is defined by $\gamma_1(G) = G$ and

$$\gamma_{n+1}(G) = [\gamma_n(G), G] \quad (n \geq 1).$$

2. The upper central series is given by $Z_0(G) = 1$ and

$$Z_n(G) = \{a \in G : [a, g_1, \dots, g_n] = 1 \text{ for all } g_1, \dots, g_n \in G\} \quad (n \geq 1).$$

Remark 2.13. This description of Z_n is equivalent to the usual recursive definition of the upper central series. Also, this form is useful for our argument.

Fact 2.14. 1. $[\gamma_i(G), \gamma_j(G)] \leq \gamma_{i+j}(G)$, hence $\gamma_i(G)/\gamma_{i+1}(G)$ is abelian.

2. If G is nilpotent of class at most n , then $\gamma_i(G) \leq Z_{n+1-i}(G)$ for every $1 \leq i \leq n$.

In what follows, we assume groups are of exponent 3 unless otherwise noted.

Fact 2.15. Let G be an exponent 3 group and $a, b, c, d \in G$. The following are well known and elementary.

1. $[a, b, c, d] = 1$; equivalently, G is nilpotent of class at most 3.
2. $\gamma_3(G) \leq Z(G)$ and $\gamma_2(G) \leq Z_2(G)$.
3. $\gamma_2(G)$ is abelian, but $Z_2(G)$ may not be abelian in general.
4. $[a, b, c] = [b, c, a] = [c, a, b]$. In particular, $[a, b, b] = 1$; equivalently, G is 2-Engel.
5. $[a, b^{-1}] = [a, b]^{-1} = [a^{-1}, b]$
6. $[a^b, a^c] = 1$ where $a^b = b^{-1}ab$. Hence the normal closure $\langle\langle a \rangle\rangle_G$ is abelian.
7. $[a, b, c] = [b, a, c]^{-1}$
8. $[a, bc] = [a, b][a, c][a, b, c]$
9. $[ab, c] = [a, c][b, c][a, b, c]^{-1}$
10. $[A, BC] \subset [A, B][A, C][A, B, C]$
11. $[AB, C] \subset [A, C][B, C][A, B, C]$

The next fact is also obtained by standard calculations of groups.

Fact 2.16. Let $\mathcal{I}$ and $\mathcal{J}$ be disjoint sets. Then $F_{\mathcal{V}_3}(\mathcal{I} \cup \mathcal{J}) = F_{\mathcal{V}_3}(\mathcal{I}) *_{\mathcal{V}_3} F_{\mathcal{V}_3}(\mathcal{J})$.

Lemma 2.17. Let $G, H \in \mathcal{V}_3$. Then natural mappings $j_G : G \rightarrow G *_{\mathcal{V}_3} H$ and $j_H : H \rightarrow G *_{\mathcal{V}_3} H$ are injective. Hence we can regard $G, H \leq G *_{\mathcal{V}_3} H$.

Proof. Define $f : G *_{\mathcal{V}_3} H \rightarrow G$ by $f(j_H(h)) = 1$ for all $h \in H$ and $f(j_G(g)) = g$ for all $g \in G$. Then f is a retraction of j_G and hence j_G is injective. By symmetry, j_H is also injective. $\square$

We recall the associated graded Lie algebra $\text{gr}(G)$ for a group G of exponent 3.

Definition 2.18. Let G be a group of exponent 3.

1. $\text{gr}_n(G) := \gamma_n(G) / \gamma_{n+1}(G)$.
2. For $a \in \gamma_n(G)$ and $b \in \gamma_m(G)$, define

$$[a\gamma_{n+1}(G), b\gamma_{m+1}(G)] := [a, b]\gamma_{n+m+1}(G).$$

Then $\text{gr}(G) = \bigoplus_i \text{gr}_i(G)$ is a graded Lie algebra. Indeed, the inclusions $[\gamma_i(G), \gamma_j(G)] \leq \gamma_{i+j}(G)$ show that the bracket is independent of the chosen representatives. Each quotient $\text{gr}_n(G)$ has exponent 3, and hence is naturally a vector space over $\mathbb{F}_3$. The remaining axioms follow from Fact 2.15.

The following are also immediate by Fact 2.15.

Lemma 2.19. Let G be a group of exponent 3.

1. $\text{gr}_4(G) = 0$.

2. $[a, b, c] = [b, c, a]$ and $[a, b, b] = 0$ for every $a, b, c \in \text{gr}(G)$.

3. $\text{gr}(G)$ is generated by $\text{gr}_1(G)$ as a Lie algebra over $\mathbb{F}_3$.

Notation 2.20. 1. For $a \in \gamma_i(G)$, we define $\text{gr}_i(a) = a\gamma_{i+1}(G) \in \text{gr}_i(G)$.

2. For $K \leq G$,

$$\text{gr}_n^G(K) := \{\text{gr}_n(a) \mid a \in \gamma_n(G) \cap K\} = \frac{(K \cap \gamma_n(G))\gamma_{n+1}(G)}{\gamma_{n+1}(G)} \leq \text{gr}_n(G).$$

$$\text{Hence } \text{gr}_n^G(K) \cong (K \cap \gamma_n(G))/(\gamma_{n+1}(G)).$$

Example 2.21. 1. If G is abelian, then $\text{gr}(G) = \text{gr}_1(G) = G$.

2. If $G = H \times K$, then $\gamma_i(G) = \gamma_i(H) \times \gamma_i(K)$ and $\text{gr}_i(G) \cong \text{gr}_i(H) \oplus \text{gr}_i(K)$.

3. Let $F_2 := F_{\mathcal{V}_3}(\{x, y\})$ be a free group in $\mathcal{V}_3$ generated by $\{x, y\}$. Then $\gamma_2(F_2) = \{[x, y]^k \mid 0 \leq k \leq 2\}$ and $\gamma_3(F_2) = 1$, hence $\text{gr}_1(F_2) = \mathbb{F}_3\bar{x} \oplus \mathbb{F}_3\bar{y}$ and $\text{gr}_2(F_2) = \mathbb{F}_3[x, y]$.

If $f : G \rightarrow H$ is a group homomorphism, then $\gamma_i(f(G)) \leq \gamma_i(H)$ holds. So we can consider an induced homomorphism between associated graded Lie algebras.

Definition 2.22. For a group homomorphism $f : G \rightarrow H$, put $\text{gr}(f) = \bigoplus_{i \geq 1} \text{gr}_i(f) : \text{gr}(G) \rightarrow \text{gr}(H)$, where

$$\text{gr}_i(f)(a\gamma_{i+1}(G)) = f(a)\gamma_{i+1}(H).$$

We can describe when $\text{gr}(f)$ is injective by using strict extensions given in the next definition.

Definition 2.23. Let $G \leq H$ be groups of exponent 3. We say that G is *strict with respect to the lower central series* in H , and write $G \leq_{\text{LCS}} H$, if

1. $\gamma_2(G) = \gamma_2(H) \cap G$ and

2. $\gamma_3(G) = \gamma_3(H) \cap G$.

Proposition 2.24. Let $f : G \rightarrow H$ be a homomorphism of exponent 3 groups.

1. If $\text{gr}(f) : \text{gr}(G) \rightarrow \text{gr}(H)$ is injective then so is f .

2. If f is injective, then the following are equivalent.

(a) $\text{gr}(f) : \text{gr}(G) \rightarrow \text{gr}(H)$ is injective.

(b) $f(G) \leq_{\text{LCS}} H$.

Proof. For (1), suppose that f is not injective and choose $1 \neq a \in \ker(f)$. Since $\gamma_4(G) = 1$, there is a largest $i \in \{1, 2, 3\}$ such that $a \in \gamma_i(G)$. Then $a\gamma_{i+1}(G)$ is a nonzero element of $\text{gr}_i(G)$ that belongs to $\ker(\text{gr}_i(f))$. Hence $\text{gr}(f)$ is not injective.

For (2), assume that f is injective and identify G with $f(G) \leq H$. For $i = 1, 2, 3$, one has

$$\ker(\text{gr}_i(f)) = \frac{\gamma_i(G) \cap \gamma_{i+1}(H)}{\gamma_{i+1}(G)}.$$

Thus $\text{gr}_1(f)$ is injective if and only if $G \cap \gamma_2(H) = \gamma_2(G)$. Under this equality, $\text{gr}_2(f)$ is injective if and only if $G \cap \gamma_3(H) = \gamma_3(G)$. Finally, $\text{gr}_3(f)$ is injective because $\gamma_4(H) = 1$. These are exactly the two conditions defining $G \leq_{\text{LCS}} H$. $\square$

Definition 2.25. Let $G \in \mathcal{V}_3$. We say that the lower central series of G coincides with the upper central series in reverse order if $\gamma_i(G) = Z_{4-i}(G)$ for every $1 \leq i \leq 3$.

Lemma 2.26. Let $G \leq H \in \mathcal{V}_3$.

1. $\gamma_i(G) \subset \gamma_i(H) \cap G \subset Z_{4-i}(H) \cap G \subset Z_{4-i}(G)$ for $1 \leq i \leq 3$.
2. If the lower central series of G coincides with the upper central series in reverse order, then $G \leq_{\text{LCS}} H$.

Proof. (1) is immediate from the definition of γ_i and Z_{4-i} , because $\gamma_4(H) = 1$.

(2) is an easy application of (1). □

To determine an associated graded Lie algebra $\text{gr}(F)$ of $F = F_{\mathcal{V}_3}(\mathcal{I})$, we recall the following result:

Fact 2.27 (Levi and van der Waerden [LW1933]). Let $\mathcal{I} = \{x_0, \dots, x_{r-1}\}$ and $F = F_{\mathcal{V}_3}(\mathcal{I}) = B(r, 3)$ the Burnside group of r -generators and exponent 3. Then $|F| = 3^{t(r)}$ where $t(r) = r + \binom{r}{2} + \binom{r}{3}$, and every $a \in F$ is uniquely expressed as follows.

$$a = x_0^{l_0} \cdots x_{r-1}^{l_{r-1}} \prod_{i < j < r} [x_i, x_j]^{m_{i,j}} \prod_{i < j < k < r} [x_i, x_j, x_k]^{n_{i,j,k}} \quad (l_i, m_{i,j}, n_{i,j,k} \in \mathbb{F}_3)$$

Remark 2.28. When $\mathcal{I} = \{x_i \mid i < \kappa\}$ for some infinite cardinal κ , then each $a \in F_{\mathcal{V}_3}(\mathcal{I})$ has a unique expression

$$a = \prod_{i < \kappa} x_i^{l_i} \prod_{i < j < \kappa} [x_i, x_j]^{m_{i,j}} \prod_{i < j < k < \kappa} [x_i, x_j, x_k]^{n_{i,j,k}}$$

where $l_i, m_{i,j}, n_{i,j,k} \in \mathbb{F}_3$ and the coefficient families (l_i) , $(m_{i,j})$, and $(n_{i,j,k})$ have finite support. This follows from the identity $F_{\mathcal{V}_3}(\mathcal{I} \sqcup \mathcal{J}) = F_{\mathcal{V}_3}(\mathcal{I}) *_{\mathcal{V}_3} F_{\mathcal{V}_3}(\mathcal{J})$.

We use this normal form to identify the associated graded Lie algebra of $B(r, 3)$.

Proposition 2.29. Let F be the Burnside group $B(r, 3) = F_{\mathcal{V}_3}(\{x_0, \dots, x_{r-1}\})$. Let V be a $\mathbb{F}_3$ -vector space with basis $\{x_0, \dots, x_{r-1}\}$. Define linear maps $\sigma_n : \text{gr}_n(F) \rightarrow \Lambda^n V$ by

1. $\overline{x_i} \mapsto x_i$ for $n = 1$,
2. $\overline{[x_i, x_j]} \mapsto x_i \wedge x_j$ for $n = 2$ and $i < j$,
3. $\overline{[x_i, x_j, x_k]} \mapsto x_i \wedge x_j \wedge x_k$ for $n = 3$ and $i < j < k$,

where $\bar{a} = a\gamma_{i+1}(F) \in \text{gr}_i(F)$ for $a \in \gamma_i(F)$. Then

$$\sigma = \sigma_1 \oplus \sigma_2 \oplus \sigma_3 : \text{gr}(F) \longrightarrow \Lambda^{[1,3]} V$$

is an isomorphism of graded Lie algebras, where the bracket on $\Lambda^{[1,3]} V$ is the one defined in Example 2.9.

Proof. By Fact 2.27, the following are bases over $\mathbb{F}_3$:

$$\begin{aligned} \{\overline{x_i} : 0 \leq i < r\} & \text{ for } \text{gr}_1(F), \\ \{\overline{[x_i, x_j]} : i < j\} & \text{ for } \text{gr}_2(F), \end{aligned}$$

and

$$\{\overline{[x_i, x_j, x_k]} : i < j < k\} \quad \text{for } \text{gr}_3(F).$$

Hence each σ_n is a linear isomorphism. The definition of the bracket gives

$$\sigma_2(\overline{[x_i, x_j]}) = x_i \wedge x_j$$

and

$$\sigma_3(\overline{[[x_i, x_j], \overline{x_k}]}) = x_i \wedge x_j \wedge x_k.$$

The brackets in the opposite order are determined by alternation, and all brackets of total degree at least 4 vanish. Thus σ preserves the Lie bracket. $\square$

Remark 2.30. The above result remains true when F is generated by an infinite set $\{x_i \mid i < \kappa\}$ for some cardinal number κ . See Remark 2.28.

Using this isomorphism, we identify $\text{gr}_i(B(r, 3))$ with $\Lambda^i V$ for $i = 1, 2, 3$.

Now let $G \in \mathcal{V}_3$ be a finite group generated by $\{g_i \mid 1 \leq i \leq r\} \subset G$. Then G is a quotient of $F_{\mathcal{V}_3}(\{x_1, \dots, x_r\})$, via the homomorphism induced by $x_i \mapsto g_i$. Hence, in order to understand $\text{gr}(G)$, we need to know how to calculate the associated graded Lie algebra $\text{gr}(F/N)$.

Now we calculate the associated graded Lie algebras of quotient groups.

Recall that

$$\text{gr}_i^G(N) = \frac{(N \cap \gamma_i(G))\gamma_{i+1}(G)}{\gamma_{i+1}(G)} \leq \text{gr}_i(G)$$

for $N \leq G$.

Lemma 2.31. *Let G be a group of exponent 3 and let $N \trianglelefteq G$. Then there is a canonical isomorphism $\text{gr}_i(G/N) \cong \text{gr}_i(G)/\text{gr}_i^G(N)$.*

Proof. Since $\gamma_i(G/N) = \gamma_i(G)N/N$,

$$\text{gr}_i(G/N) \cong \frac{\gamma_i(G)N}{\gamma_{i+1}(G)N} \cong \frac{\gamma_i(G)}{\gamma_i(G) \cap (\gamma_{i+1}(G)N)} = \frac{\gamma_i(G)}{\gamma_{i+1}(G)(\gamma_i(G) \cap N)} \cong \frac{\text{gr}_i(G)}{\text{gr}_i^G(N)}.$$

$\square$

3 Main results

In this section we prove that T_3 has a model companion, assuming Proposition A, which is proved in the next section.

Proposition 3.1. *For any $a \in G \in \mathcal{V}_3$,*

$$\langle\langle a \rangle\rangle_G = \{a^{g_0} \cdots a^{g_{n-1}} \mid g_i \in G, n \leq 3\}.$$

In other words, every element in $\langle\langle a \rangle\rangle_G$ is a product of at most 3 conjugates of a .

Proof. Let $E_a = \{a^k[a, g] \mid g \in G, k \in \mathbb{F}_3\} \subset \langle\langle a \rangle\rangle$. Because $[a, g][a, g'] = [a, gg'[g, g']]$, $[a, g]^{-1} = [a, g^{-1}]$ and $a[a, g] = [a, g]a$, E_a is a subgroup. Since $a^g = g^{-1}ag = a[a^{-1}, g^{-1}] = a[a, g]$, $E_a = \langle\langle a \rangle\rangle_G$. Since $a[a, g] = a^g$, $a^2[a, g] = aa^g$, and $[a, g] = a^{-1}a^g = a^2a^g$, every element in E_a is a product of at most 3 conjugates of a . $\square$

Lemma 3.2. *Let $B, G \leq H \in \mathcal{V}_3$, $d(B) \leq m$, $\langle B, G \rangle = H$ and $\Delta \subset H$ with $|\Delta| \leq n$. Then for any $h \in \langle\langle \Delta \rangle\rangle_H$, there is $C \leq G$ such that*

1. $d(C) \leq 3(m+1)n$,
2. $h \in \langle\langle \Delta \rangle\rangle_{H_0}$ where $H_0 = \langle C, B, \Delta \rangle \leq H$.

Proof. Let $\Delta = \{\delta_i \mid i < r\}$ with $r \leq n$. Then $\langle\langle \Delta \rangle\rangle_H = \langle\langle \delta_0 \rangle\rangle_H \cdots \langle\langle \delta_{r-1} \rangle\rangle_H$, so it suffices to prove the claim for $\Delta = \{\delta\}$.

By Proposition 3.1, $h = \prod_{i < k} \delta^{u_i}$ for some $u_i \in H$ and $k \leq 3$. Since $H = \langle G, B \rangle$, every $u \in H$ is expressed as

$$u = gb \prod_{j < m} [g_j, b_j]^{k_j} \mod \gamma_3(H)$$

for some $g, g_j \in G$, $b \in B$ and $k_j \in \mathbb{F}_3$ where $B = \langle b_j \mid j < m \rangle$. Here, we used that $[gg', b] = [g, b][g', b][g, g', b]^{-1}$, $[b, g] = [g, b]^{-1}$ and $[H, H]$ is abelian. Because $\gamma_3(H) \leq Z(H)$, $\delta^u = \delta^v$ where $u = v \mod \gamma_3(H)$.

Hence C is obtained by collecting all $g, g_j \in G$ in the above expression for each u_i ($i < k$) as generators of C .

This argument shows that C is generated by at most $k(m+1)|\Delta| \leq k(m+1)n \leq 3(m+1)n$ elements. $\square$

A proof of the next proposition is given in the next section as Proposition 4.13.

Proposition A. *There is a function $f_0 : \mathbb{N} \rightarrow \mathbb{N}$ such that for every $n \in \mathbb{N}$, every e.c. model M of T_3 , and every $A \leq M$ with $d(A) \leq n$, there is $A \leq D \leq_{\text{LCS}} M$ such that D is generated by at most $f_0(n)$ elements.*

Theorem 3.3 (Main theorem). *There is a function $f : \mathbb{N} \rightarrow \mathbb{N}$ satisfying the following: Let T_3 be the theory of groups of exponent 3 and M an existentially closed model of T_3 . Let $A \leq M$ be a finite subgroup and let $A \leq B \in \mathcal{V}_3$ be such that $d(B) \leq m$. If M and B do not have an amalgam in $\mathcal{V}_3$ over A , then there is $A \leq D \leq M$ such that*

1. D and B do not have an amalgam in $\mathcal{V}_3$ over A .
2. $d(D) \leq f(m)$.

Proof. Suppose that $A \leq M$ and $A \leq B$ do not have an amalgam over A . Let $H = M *_{\mathcal{V}_3} B$ and let $j_M : M \rightarrow H$ and $j_B : B \rightarrow H$ be canonical embeddings. Let $d(A) = n$ and let A be generated by $\{a_0, \dots, a_{n-1}\}$. Put $\Delta = \{\delta_{a_i} \mid i < n\} \subset H$ where $\delta_a = j_M(a)j_B(a)^{-1}$. By the assumption, at least one of the canonical homomorphisms $\bar{j}_M : M \rightarrow H/\langle\langle \Delta \rangle\rangle_H$ and $\bar{j}_B : B \rightarrow H/\langle\langle \Delta \rangle\rangle_H$ is not injective; equivalently, at least one of the following holds:

- There is $1 \neq g \in M$ such that $j_M(g) \in \langle\langle \Delta \rangle\rangle_H$.
- There is $1 \neq g \in B$ such that $j_B(g) \in \langle\langle \Delta \rangle\rangle_H$.

Since the other case is similar, we assume $g \in M$ and $j_M(g) \in \langle\langle \Delta \rangle\rangle_H$.

Recall that $d(A) = n$ and $d(B) \leq m$. By Lemma 3.2, there is $C_0 \leq M$ generated by $3(m+1)n$ elements such that, by letting $H_0 = \langle j_M(C_0), j_B(B), \Delta \rangle \leq H$, $j_M(g) \in \langle\langle \Delta \rangle\rangle_{H_0}$. Let $C = \langle C_0, A, g \rangle_M$, which is generated by $3(m+1)n + n + 1$ elements. Since $n = d(A) \leq \log_3(|A|) \leq \log_3(|B|) \leq t(m)$ where $t(m) = m + \binom{m}{2} + \binom{m}{3}$, we have $3(m+1)n + n + 1 \leq (3m+4)t(m) + 1$. By Proposition A, we can find $C \leq D \leq_{\text{LCS}} M$ such that D is generated by at most $f(m) := f_0((3m+4)t(m) + 1)$ elements.

Claim A. D and B do not have an amalgam in $\mathcal{V}_3$ over A .

Since $D \leq_{\text{LCS}} M$, a natural homomorphism $D *_{\mathcal{V}_3} B \rightarrow M *_{\mathcal{V}_3} B$ is an embedding, so we identify $D *_{\mathcal{V}_3} B$ as a subgroup of $M *_{\mathcal{V}_3} B$. Since $j_M|_D$ agrees with the natural embedding $j_D : D \rightarrow D *_{\mathcal{V}_3} B \leq M *_{\mathcal{V}_3} B$ and $H_0 \leq \langle j_D(D), j_B(B) \rangle \leq D *_{\mathcal{V}_3} B$, we know $1 \neq j_D(g) = j_M(g) \in \langle\langle \Delta \rangle\rangle_{D *_{\mathcal{V}_3} B} \leq D *_{\mathcal{V}_3} B$. Hence $D \rightarrow D *_{\mathcal{V}_3} B / \langle\langle \Delta \rangle\rangle_{D *_{\mathcal{V}_3} B}$ is not an embedding. This shows that $A \leq D$ and $A \leq B$ do not have an amalgam over A . $\square$

By Fact 2.6, we have the following.

Corollary 3.4. *Let T_3 be the theory of groups of exponent 3. Then T_3 has a model companion.*

4 Structural analysis

We first construct a relatively free presentation of G .

Let G be a group of exponent 3. Since $V_G = G/\gamma_2(G)$ is abelian, it is a vector space over $\mathbb{F}_3$. Let $\mathcal{I} = \{e_i \mid i \in I\}$ be a basis of V_G . For each $i \in I$, choose a representative $a_i \in G$ such that $e_i = a_i\gamma_2(G)$.

Proposition 4.1. *The assignment $e_i \mapsto a_i$ induces a surjective homomorphism $\varphi_G : F_{\mathcal{V}_3}(\mathcal{I}) \rightarrow G$ such that $\ker(\varphi_G) \leq \gamma_2(F_{\mathcal{V}_3}(\mathcal{I}))$.*

Proof. Let $F = F_{\mathcal{V}_3}(\mathcal{I})$ and $H = \langle a_i \mid i \in I \rangle \leq G$. Since the cosets $a_i\gamma_2(G)$ form a basis of $G/\gamma_2(G)$, we have $G = H\gamma_2(G)$.

We claim that $H = G$. Since $\gamma_2(G)$ is abelian and $\gamma_3(G) \leq Z(G)$, the commutator identities give

$$\gamma_2(G) = [H\gamma_2(G), H\gamma_2(G)] \leq H[H, \gamma_2(G)] \leq H\gamma_3(G).$$

Consequently,

$$[H, \gamma_2(G)] \leq [H, H\gamma_3(G)] = [H, H] \leq H,$$

where we used the centrality of $\gamma_3(G)$. It follows that $\gamma_2(G) \leq H[H, \gamma_2(G)] \leq H$, and hence $G = H\gamma_2(G) = H$. Thus φ_G is surjective.

The induced map

$$\bar{\varphi}_G : F/\gamma_2(F) \longrightarrow G/\gamma_2(G)$$

sends the basis $\mathcal{I}$ bijectively to the basis $\{a_i\gamma_2(G) : i \in I\}$, so it is an isomorphism. If $a \in \ker(\varphi_G)$, then $a\gamma_2(F) \in \ker(\bar{\varphi}_G) = 0$. Therefore $a \in \gamma_2(F)$, as required. $\square$

Remark 4.2. The above argument shows that, in general, if $H \leq G$ and $G = H\gamma_2(G)$, then $G = H$.

Lemma 4.3. *Let G be a group of exponent 3, let $K \leq \gamma_2(G)$, and let $L = \langle\langle K \rangle\rangle_G$ be the normal closure of K in G . Then*

1. $L = K[K, G]$ and $L \leq \gamma_2(G)$,
2. $\text{gr}_2^G(L) = \text{gr}_2^G(K) = \{\text{gr}_2(a) \in \text{gr}_2(G) \mid a \in K \cap \gamma_2(G)\}$,
3. $\text{gr}_3^G(L) = \text{gr}_3^G(K) + [\text{gr}_2^G(K), \text{gr}_1(G)]$.

Proof. Since $K \leq \gamma_2(G)$, we have $[K, G] \leq \gamma_3(G) \leq Z(G)$. For $k \in K$ and $g \in G$,

$$k^g = [g^{-1}, k]k \in K[K, G].$$

Thus every conjugate of an element of K belongs to $K[K, G]$. The reverse inclusion follows because the normal closure of K contains both K and $[K, G]$. Hence

$$L = K[K, G] \leq \gamma_2(G).$$

Since $[K, G] \leq \gamma_3(G)$, the images of L and K in $\text{gr}_2(G) = \gamma_2(G)/\gamma_3(G)$ coincide. This proves (2).

Moreover,

$$L \cap \gamma_3(G) = (K \cap \gamma_3(G))[K, G].$$

For $k \in K$ and $g \in G$, the induced bracket is

$$[k\gamma_3(G), g\gamma_2(G)] = [k, g]\gamma_4(G) = [k, g].$$

Therefore the image of $[K, G]$ in $\text{gr}_3(G)$ is exactly $[\text{gr}_2^G(K), \text{gr}_1(G)]$, which proves (3). $\square$

The next proposition is a main tool to analyze group structures of exponent 3 in this section.

Proposition 4.4. *Let G_i ($i = 0, 1$) be groups of exponent 3.*

1. *Then the following natural maps induced by the canonical embeddings $G_i \rightarrow G_0 *_{\mathcal{V}_3} G_1$ ($i = 0, 1$) are isomorphisms.*

$$\eta_1 : \text{gr}_1(G_0) \oplus \text{gr}_1(G_1) \rightarrow \text{gr}_1(G_0 *_{\mathcal{V}_3} G_1)$$

$$\eta_2 : \text{gr}_2(G_0) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_1(G_1)) \oplus \text{gr}_2(G_1) \rightarrow \text{gr}_2(G_0 *_{\mathcal{V}_3} G_1)$$

$$\eta_3 : \text{gr}_3(G_0) \oplus (\text{gr}_2(G_0) \otimes \text{gr}_1(G_1)) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_2(G_1)) \oplus \text{gr}_3(G_1) \rightarrow \text{gr}_3(G_0 *_{\mathcal{V}_3} G_1)$$

$$\text{where } \text{gr}_i(G_0) \otimes \text{gr}_j(G_1) \ni a \otimes b \xrightarrow{\eta_{i+j}} [\eta_i(a), \eta_j(b)] \in \text{gr}_{i+j}(G_0 *_{\mathcal{V}_3} G_1).$$

2. *Let us identify $\text{dom}(\eta_i)$ with $\text{Im}(\eta_i)$. Then, by letting $\mathcal{B}_{i,j} = \text{gr}_i(G_0) \otimes \text{gr}_j(G_1)$ for $i, j \geq 1$ with $i + j \leq 3$, $\mathcal{B}_{i,0} = \text{gr}_i(G_0)$, $\mathcal{B}_{0,j} = \text{gr}_j(G_1)$, and $\mathcal{B}_{i,j} = 0$ ($i + j \geq 4$),*

$$[\mathcal{B}_{i,j}, \mathcal{B}_{k,l}] \subset \mathcal{B}_{i+k,j+l}.$$

Proof. (2) is immediate from the definition of η_i . We show (1). Let $V_n = \text{gr}_1(G_n)$ ($n < 2$) and fix a basis $\mathcal{I}_n$ of V_n and a representative $a_e \in G_n$ of each vector $e \in \mathcal{I}_n$. Let us consider a lift $F_{G_n} := F_{\mathcal{V}_3}(\mathcal{I}_n)$ of G_n , and we will use the chosen surjective homomorphism $\varphi_n : F_{G_n} \rightarrow G_n$ and its kernel $K_n := \ker(\varphi_n) \leq \gamma_2(F_{G_n})$ as defined in Proposition 4.1. Let $V = V_0 \oplus V_1$, $F = F_{G_0} *_{\mathcal{V}_3} F_{G_1} = F_{\mathcal{V}_3}(\mathcal{I}_0 \sqcup \mathcal{I}_1)$, $K = \langle K_0, K_1 \rangle_F = K_0 K_1$, and $L = \langle\langle K \rangle\rangle_F$. (Notice that $\text{gr}_1(F) = V$.) Then we can identify $G_0 *_{\mathcal{V}_3} G_1$ with F/L .

By Lemma 2.31, we have a canonical isomorphism $\text{gr}_i(F/L) \cong \text{gr}_i(F)/\text{gr}_i^F(L)$. Also by Lemma 4.3,

$$\text{gr}_1^F(L) = 0, \quad \text{gr}_2^F(L) = \text{gr}_2^F(K), \quad \text{gr}_3^F(L) = \text{gr}_3^F(K) + [\text{gr}_2^F(K), \text{gr}_1^F(F)].$$

Now recall that we can identify $\text{gr}_i(F)$ with $\Lambda^i V$ and recall the standard direct sum decomposition of Grassmann algebra:

$$\begin{aligned}\Lambda^1(V_0 \oplus V_1) &= \Lambda^1 V_0 \oplus \Lambda^1 V_1 = V_0 \oplus V_1, \\ \Lambda^2(V_0 \oplus V_1) &= \Lambda^2 V_0 \oplus (V_0 \otimes V_1) \oplus \Lambda^2 V_1, \\ \Lambda^3(V_0 \oplus V_1) &= \Lambda^3 V_0 \oplus (\Lambda^2 V_0 \otimes V_1) \oplus (V_0 \otimes \Lambda^2 V_1) \oplus \Lambda^3 V_1.\end{aligned}$$

Hence,

$$\begin{aligned}\text{gr}_1(G_0 *_{\mathcal{V}_3} G_1) &\cong \text{gr}_1(F) / \text{gr}_1^F(L) \\ &= V_0 \oplus V_1.\end{aligned}$$

Since $K = K_0 K_1$ is abelian, by letting $\text{gr}_2^F(K_n) = R_n \subset \Lambda^2 V_n$,

$$\text{gr}_2^F(K) = R_0 \oplus R_1 \subset \Lambda^2 V_0 \oplus \Lambda^2 V_1 \subset \Lambda^2 V.$$

Hence $\text{gr}_2^F(K)$ is block-homogeneous with respect to the decomposition. After we identify $\text{gr}_2(F_{G_n})$ with $\Lambda^2 V_n$, we can consider $\text{gr}_2^{F_{G_n}}(K_n) = \text{gr}_2^F(K_n) = R_n$. Hence, because $\Lambda^2 V_n / R_n \cong \text{gr}_2(F_{G_n} / K_n) = \text{gr}_2(G_n)$,

$$\begin{aligned}\text{gr}_2(G_0 *_{\mathcal{V}_3} G_1) &\cong \text{gr}_2(F) / \text{gr}_2^F(L) \\ &= \Lambda^2(V_0 \oplus V_1) / \text{gr}_2^F(K) \\ &\cong \frac{\Lambda^2 V_0 \oplus (V_0 \otimes V_1) \oplus \Lambda^2 V_1}{R_0 \oplus R_1} \\ &\cong \text{gr}_2(G_0) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_1(G_1)) \oplus \text{gr}_2(G_1).\end{aligned}$$

Let $S = \text{gr}_3^F(L) \subset \Lambda^3 V$, $S_n = \text{gr}_3^{F_{G_n}}(K_n) \subset \Lambda^3 V_n$. Then $\text{gr}_3^F(K) = S_0 \oplus S_1$. Arguing as above,

$$S = S_0 + S_1 + [R_0 + R_1, V] = S_0 + S_1 + (R_0 + R_1) \wedge (V_0 + V_1).$$

Because $R_n \wedge V_n \subset S_n$, we have

$$S = S_0 + R_0 \wedge V_1 + V_0 \wedge R_1 + S_1, \quad R_n \wedge V_{1-n} \cong R_n \otimes V_{1-n}.$$

Hence,

$$\begin{aligned}\text{gr}_3(G_0 *_{\mathcal{V}_3} G_1) &\cong \text{gr}_3(F) / \text{gr}_3^F(L) \\ &\cong \frac{\Lambda^3(V_0) \oplus (\Lambda^2 V_0 \otimes V_1) \oplus (V_0 \otimes \Lambda^2 V_1) \oplus \Lambda^3 V_1}{S_0 + R_0 \otimes V_1 + V_0 \otimes R_1 + S_1} \\ &\cong \text{gr}_3(G_0) \oplus (\text{gr}_2(G_0) \otimes \text{gr}_1(G_1)) \oplus (\text{gr}_1(G_0) \otimes \text{gr}_2(G_1)) \oplus \text{gr}_3(G_1),\end{aligned}$$

because $S = S_0 + R_0 \otimes V_1 + V_0 \otimes R_1 + S_1$ is block-homogeneous. $\square$

In the remainder of this section, we will often consider $H = G *_{\mathcal{V}_3} F_n$ where $F_n = F_{\mathcal{V}_3}(x_1, \dots, x_n)$. By Proposition 4.4, we use the following identifications:

$$\begin{aligned}\text{gr}_1(H) &= \text{gr}_1(G) \oplus \text{gr}_1(F_n) \\ \text{gr}_2(H) &= \text{gr}_2(G) \oplus (\text{gr}_1(G) \otimes \text{gr}_1(F_n)) \oplus \text{gr}_2(F_n) \\ \text{gr}_3(H) &= \text{gr}_3(G) \oplus (\text{gr}_2(G) \otimes \text{gr}_1(F_n)) \oplus (\text{gr}_1(G) \otimes \text{gr}_2(F_n)) \oplus \text{gr}_3(F_n)\end{aligned}$$

and, by Proposition 2.29,

$$\text{gr}_i F_n = \Lambda^i V_n$$

where V_n is a vector space over $\mathbb{F}_3$ with basis $\{x_j \mid 1 \leq j \leq n\}$.

Lemma 4.5. *If $D \leq_{\text{LCS}} G \in \mathcal{V}_3$, then for every $B \in \mathcal{V}_3$, a natural homomorphism $D *_{\mathcal{V}_3} B \rightarrow G *_{\mathcal{V}_3} B$ induced by the inclusion $D \leq G$ is injective.*

Proof. Since $D \leq_{\text{LCS}} G$, a natural homomorphism $\text{gr}(D) \rightarrow \text{gr}(G)$ is injective by Proposition 2.24. Hence, by Proposition 4.4, $\text{gr}(D *_{\mathcal{V}_3} B) \rightarrow \text{gr}(G *_{\mathcal{V}_3} B)$ is injective. Again, by Proposition 2.24, a natural homomorphism $D *_{\mathcal{V}_3} B \rightarrow G *_{\mathcal{V}_3} B$ is injective. $\square$

Lemma 4.6. *Let $1 \neq G \in \mathcal{V}_3$ and let $F_2 = F_{\mathcal{V}_3}(x, y)$ be the relatively free group of rank 2 in $\mathcal{V}_3$. Then $G \leq_{\text{LCS}} G *_{\mathcal{V}_3} F_2$, and the lower central series of $G *_{\mathcal{V}_3} F_2$ coincides with its upper central series in reverse order.*

Proof. Let $H = G *_{\mathcal{V}_3} F_2$. Let $f : G \rightarrow H$ be a natural inclusion map. By Proposition 4.4, $\text{gr}(f) : \text{gr} G \rightarrow \text{gr} H$ is injective. Hence $G \leq_{\text{LCS}} H$ by Proposition 2.24.

Claim A. $\gamma_2(H) = Z_2(H)$.

Let $a \in H \setminus \gamma_2(H)$. We show that there is $b, c \in H$ such that $[a, b, c] \neq 1$, which implies $a \notin Z_2(H)$. Because $\text{gr}_1(H) = \text{gr}_1(G) \oplus \text{gr}_1(F_2)$, $\text{gr}_1(a) = \alpha + \beta$ for some $\alpha \in \text{gr}_1(G)$ and $\beta \in \text{gr}_1(F_2) = \Lambda^1 V_2$ where V_2 is a $\mathbb{F}_3$ -vector space with a basis $\{x, y\}$.

Case 1: Suppose $\alpha \neq 0$. Then $\text{gr}_3([a, x, y]) = -\alpha \otimes (x \wedge y) + \beta \wedge x \wedge y \neq 0$, hence $[a, x, y] \neq 1$.

Case 2: Suppose $\alpha = 0$ and $\beta \neq 0$. Take $g \in G \setminus \gamma_2(G)$. Since $\dim \text{gr}_1(F_2) = 2$, we can find $w \in F_2$ such that $\{\beta, \text{gr}_1(w)\}$ is linearly independent in $\text{gr}_1(F_2)$. Then $[a, g, w] \neq 1$ because $\text{gr}_3([a, g, w]) = \text{gr}_1(g) \otimes (\beta \wedge \text{gr}_1(w)) \neq 0$.

Claim B. $\gamma_3(H) = Z(H)$

Let $a \notin \gamma_3(H)$. We show that there is $b \in H$ such that $[a, b] \neq 1$, which implies $a \notin Z(H)$. Because of Claim A, we can assume $a \in \gamma_2(H)$. Since $\text{gr}_2(H) = \text{gr}_2(G) \oplus (\text{gr}_1(G) \otimes \text{gr}_1(F_2)) \oplus \text{gr}_2(F_2)$, we have $\text{gr}_2(a) = \alpha_2 + \alpha_1 \otimes x + \alpha'_1 \otimes y + \beta_2$ for some $\alpha_2 \in \text{gr}_2(G)$, $\alpha_1, \alpha'_1 \in \text{gr}_1(G)$, and $\beta_2 \in \text{gr}_2(F_2)$. (Here, we used that $\text{gr}_1(F_2) = \mathbb{F}_3 x \oplus \mathbb{F}_3 y$.)

Case 1: If $\alpha_2 \neq 0$, then $[a, x] \neq 1$.

Case 2: If $\alpha_2 = 0$ and $\alpha_1 \neq 0$, then $\text{gr}_3([a, y]) = -\alpha_1 \otimes (x \wedge y) + \beta_2 \wedge y = -\alpha_1 \otimes (x \wedge y) \neq 0$. In the case of $\alpha'_1 \neq 0$, use x instead of y .

Case 3: If $\alpha_2 = 0$, $\alpha_1 = \alpha'_1 = 0$ and $\beta_2 \neq 0$, then choose $g \in G \setminus \gamma_2(G)$. We have $[a, g] \neq 1$ because $\text{gr}_3([a, g]) = -\text{gr}_1(g) \otimes \beta_2 \neq 0$. $\square$

Lemma 4.7. *Let $G \in \mathcal{V}_3$, $g_1, \dots, g_n \in \gamma_2(G)$ and $F_{2n} = F_{\mathcal{V}_3}(x_1, y_1, \dots, x_n, y_n)$. Then a natural homomorphism $G \rightarrow H$ is injective where*

$$H = (G *_{\mathcal{V}_3} F_{2n}) / \langle\langle g_i^{-1}[x_i, y_i] \mid 1 \leq i \leq n \rangle\rangle.$$

Proof. Let

$$H_0 = G *_{\mathcal{V}_3} F_{2n}, \quad r_i = g_i^{-1}[x_i, y_i] \in H_0$$

$$K = \langle r_i \mid i \leq n \rangle_{H_0} \leq \gamma_2(H_0), \quad L = \langle\langle K \rangle\rangle_{H_0} = \langle\langle r_1 \rangle\rangle_{H_0} \cdots \langle\langle r_n \rangle\rangle_{H_0}, \quad H = H_0/L.$$

Claim A. $L \leq \gamma_2(H_0)$ and $\langle\langle r_i \rangle\rangle = \{r_i^m[r_i, h] \mid m \in \mathbb{F}_3, h \in H_0\}$.

Since $L = K[K, H_0]$ and $K \leq \gamma_2(H_0)$, we have $L \leq \gamma_2(H_0)$. Similarly,

$$\langle\langle r_i \rangle\rangle = \langle r_i \rangle[\langle r_i \rangle, H_0] = \langle r_i \rangle[r_i, H_0].$$

Here, we used that $[r^{-1}, h] = [r, h]^{-1} = [r, h^{-1}]$. Since $r_i \in \gamma_2(H_0)$, $[r_i, h][r_i, h'] = [r_i, hh']$. Hence $[r_i, H_0] = \{[r_i, h] \mid h \in H_0\}$.

It suffices to show that $L \cap G = 1$, so let $a \in L \cap G$. By Claim A, $a = \prod_i r_i^{m_i}[r_i, h_i] \in \gamma_2(H_0)$ for some $h_i \in H_0$.

Claim B. $m_i = 0$ ($i \leq n$). Hence $a = \prod_i [r_i, h_i] \in \gamma_3(H_0)$.

Since $g_i, r_i, a \in \gamma_2(H_0)$, we can calculate that

$$\begin{aligned} \text{gr}_2(a) &= \sum_i m_i \text{gr}_2(r_i) \\ &= \sum_i m_i (\text{gr}_2([x_i, y_i]) - \text{gr}_2(g_i)) \\ &= \sum_i m_i (x_i \wedge y_i) - \sum_i m_i \text{gr}_2(g_i). \end{aligned}$$

Because $a \in G$, $\text{gr}_2(a)$ must be in $\text{gr}_2(G)$ whereas $0 \neq x_i \wedge y_i \in \text{gr}_2(F_{2n})$ and $\{x_i \wedge y_i \mid i \leq n\}$ is linearly independent. Hence we have $m_i = 0$ for all i and it follows that $a \in \gamma_3(H_0)$.

Claim C. $a = 1$.

Recall that $a = \prod_i [r_i, h_i] \in \gamma_3(H_0)$. Suppose $\text{gr}_1(h_i) = s_i + t_i$ where $s_i \in \text{gr}_1(G)$ and $t_i \in \text{gr}_1(F_{2n})$. Then,

$$\begin{aligned} \text{gr}_3(a) &= \sum_i [x_i \wedge y_i - \text{gr}_2(g_i), s_i + t_i] \\ &= \sum_i -s_i \otimes (x_i \wedge y_i) + x_i \wedge y_i \wedge t_i - [\text{gr}_2(g_i), s_i] - \text{gr}_2(g_i) \otimes t_i. \end{aligned}$$

Because $a \in G$ and $\text{gr}_3(a) \in \text{gr}_3(G)$, $\text{gr}_3(a)$ must be $-\sum_i [\text{gr}_2(g_i), s_i]$. Hence $\sum_i s_i \otimes (x_i \wedge y_i) = 0$, which implies $s_i = 0$. Therefore we have $\text{gr}_3(a) = 0$ and $a = 1$. $\square$

Lemma 4.8. *Let $C \leq G \in \mathcal{V}_3$ such that C is generated by m elements. Then there is an extension $G \leq H \in \mathcal{V}_3$ and $C \leq D \leq H$ such that*

1. D is generated by C and at most $2m$ elements in H , and
2. $D \cap \gamma_2(H) = \gamma_2(D)$.

Proof. Let $g_1, \dots, g_n \in C \cap \gamma_2(G)$ be such that $\{g_i \gamma_2(C) \mid i \leq n\}$ is a basis of $(C \cap \gamma_2(G))/\gamma_2(C)$. Then $C \cap \gamma_2(G) \subset \langle g_i \mid i \leq n \rangle \gamma_2(C)$ and $n \leq \dim \text{gr}_1(C) \leq m$. Let $F_{2n} = F_{\mathcal{V}_3}(x_1, y_1, \dots, x_n, y_n)$, $H_0 = G *_{\mathcal{V}_3} F_{2n}$ and $H = H_0/L$ where $L = \langle\langle g_i^{-1}[x_i, y_i] \mid i \leq n \rangle\rangle$. Then, by Lemma 4.7, $G \leq H$.

Claim A. $D = \langle C, x_i, y_i \mid i \leq n \rangle_H \leq H$ satisfies $D \cap \gamma_2(H) = \gamma_2(D)$.

First notice that, since $L \leq \gamma_2(H_0)$, $\text{gr}_1(H) = \text{gr}_1(H_0/L) \cong \text{gr}_1(H_0) = \text{gr}_1(G) \oplus \text{gr}_1(F_{2n})$.

Let $a \in D \cap \gamma_2(H)$. Then $a \in c \prod x_i^{m_i} y_i^{n_i} \gamma_2(D)$ for some $c \in C$. Hence, $0 = \text{gr}_1(a) = \text{gr}_1(c) + \sum_i m_i x_i + \sum_i n_i y_i$. Because $\{x_i, y_i \mid i \leq n\}$ is a basis of $\text{gr}_1(F_{2n})$, $m_i = n_i = 0$ for all $i \leq n$ and $\text{gr}_1(c) = 0$. Therefore $c \in \gamma_2(G)$ follows. This argument shows that $D \cap \gamma_2(H) = (C \cap \gamma_2(G))\gamma_2(D)$. By the choice of g_i ($i \leq n$),

$$\begin{aligned} D \cap \gamma_2(H) &= (C \cap \gamma_2(G))\gamma_2(D) \\ &\subset (\langle g_i \mid i \leq n \rangle \gamma_2(C)) \gamma_2(D) \\ &\subset \gamma_2(D). \end{aligned}$$

□

Lemma 4.9. *Let $G \in \mathcal{V}_3$, $g_1, \dots, g_n \in Z(G)$ and let $F_{3n} = F_{\mathcal{V}_3}(x_1, y_1, z_1, \dots, x_n, y_n, z_n)$. Then the natural map $G \rightarrow H$ is injective where*

$$H = (G \times F_{3n}) / \langle\langle g_i^{-1}[x_i, y_i, z_i] \mid i \leq n \rangle\rangle$$

Proof. Let $L = \langle\langle g_i^{-1}[x_i, y_i, z_i] \mid i \leq n \rangle\rangle \leq G \times F_{3n}$. Since $g_i \in Z(G)$, $g_i \in Z(G \times F_{3n})$. Hence $L = \langle g_i^{-1}[x_i, y_i, z_i] \mid i \leq n \rangle$.

We prove that $G \cap L = 1$ in $G \times F_{3n}$. Let $a \in G \cap L$. Then $a = \prod_i g_i^{-m_i} [x_i, y_i, z_i]^{m_i}$. Since $a \in G$, $m_i = 0$ for all $i \leq n$. This means $a = 1$. □

Lemma 4.10. *Let $C \leq G \in \mathcal{V}_3$ and let C be generated by at most n elements. Suppose that $C \cap \gamma_2(G) = \gamma_2(C)$. Then there are $G \leq H$ and $C \leq D \leq H$ such that*

1. D is generated by C and at most $3n(n-1)/2$ elements in H ,
2. $D \leq_{\text{LCS}} H$.

Proof. Let $g_0, \dots, g_{m-1} \in C \cap \gamma_3(G)$ be such that $\{g_i \gamma_3(C) \mid i < m\}$ generates the $\mathbb{F}_3$ -vector space $(C \cap \gamma_3(G))/\gamma_3(C)$. Since $C \cap \gamma_3(G) \leq C \cap \gamma_2(G) = \gamma_2(C)$, we can choose $m \leq \dim \text{gr}_2(C) \leq \binom{n}{2}$. By Lemma 4.9, $H = (G \times F_{3m})/L$ is an extension of G where $L = \langle g_i^{-1}[x_i, y_i, z_i] \mid i < m \rangle$. Let $D = \langle C, x_i, y_i, z_i \mid i < m \rangle_H$. Then $D = (C \times F_{3m})/L$.

Claim A. $\gamma_2(H) \cap D = \gamma_2(D)$

Since $g_i \in C \cap \gamma_3(G) \leq \gamma_2(C)$, $L \leq \gamma_2(C \times F_{3m})$. Also, $\gamma_2(H) = (\gamma_2(G \times F_{3m})L)/L = \gamma_2(G \times F_{3m})/L = (\gamma_2(G) \times \gamma_2(F_{3m}))/L$. Hence

$$\begin{aligned} \gamma_2(H) \cap D &= \frac{\gamma_2(G) \times \gamma_2(F_{3m})}{L} \cap \frac{C \times F_{3m}}{L} \\ &= \frac{(\gamma_2(G) \cap C) \times \gamma_2(F_{3m})}{L} \\ &= \frac{\gamma_2(C) \times \gamma_2(F_{3m})}{L} \\ &= \frac{\gamma_2(C \times F_{3m})}{L} \\ &= \gamma_2(D). \end{aligned}$$

Claim B. $\gamma_3(H) \cap D = \gamma_3(D)$

Since $g_i \in \gamma_3(G) \cap C$ and $[x_i, y_i, z_i] \in \gamma_3(F_{3m})$, $L \leq (\gamma_3(G) \cap C) \times \gamma_3(F_{3m}) \leq \gamma_3(G \times F_{3m})$. Hence,

$$\begin{aligned}\gamma_3(H) &= \frac{\gamma_3(G \times F_{3m})}{L}, \\ \gamma_3(H) \cap D &= \frac{(\gamma_3(G) \cap C) \times \gamma_3(F_{3m})}{L}, \\ \gamma_3(D) &= \frac{(\gamma_3(C) \times \gamma_3(F_{3m}))L}{L}.\end{aligned}$$

Suppose $a \in (\gamma_3(G) \cap C) \times \gamma_3(F_{3m})$. By the choice of g_i ($i < m$), $a = h \prod_i g_i^{m_i} f$ for some $h \in \gamma_3(C)$, $m_i \in \mathbb{F}_3$, and $f \in \gamma_3(F_{3m})$. Therefore, $a \in h(\prod_i [x_i, y_i, z_i]^{m_i})fL$. This shows that $\gamma_3(H) \cap D \subset \gamma_3(D)$. $\square$

Remark 4.11. The bound on the number of additional generators in the above lemma is not optimal. Indeed, for example, $F_{\mathcal{V}_3}(x_1, \dots, x_4)$ contains $\binom{4}{3} = 4$ elements $[x_i, x_j, x_k]$ ($i < j < k \leq 4$) that are linearly independent in $\text{gr}_3(F_4)$. Hence, for $g_1, \dots, g_4$, $L = \langle g_1^{-1}[x_1, x_2, x_3], \dots, g_4^{-1}[x_2, x_3, x_4] \rangle$ is enough to prove the desired conditions. This trick works for Lemma 4.8.

Proposition 4.12. *Let M be an e.c. model of T_3 . Then the lower central series of M coincides with its upper central series in reverse order. Moreover,*

$$\gamma_2(M) = \{[a, b] \mid a, b \in M\},$$

$$\gamma_3(M) = \{[a, b, c] \mid a, b, c \in M\}.$$

Proof. Let $a \in \gamma_2(M)$. By Lemma 4.7, there is an extension $M \leq H$ and $b', c' \in H$ such that $[b', c'] = a$. Since M is e.c., we can find $b, c \in M$ such that $[b, c] = a$ in M . This proves $\gamma_2(M) = \{[a, b] \mid a, b \in M\}$. A similar argument, using Lemma 4.9, proves the corresponding equality for $\gamma_3(M)$.

Let $a \in M \setminus \gamma_2(M)$ and let $H = M *_{\mathcal{V}_3} F_2$. Since $M \leq_{\text{LCS}} H$, $a \notin \gamma_2(H)$. Hence there are $b', c' \in H$ such that $[a, b', c'] \neq 1$. Since M is an e.c. model, we can find $b, c \in M$ such that $[a, b, c] \neq 1$, which proves $\gamma_2(M) = Z_2(M)$. A similar argument proves $\gamma_3(M) = Z(M)$. $\square$

Proposition 4.13. *Let M be an e.c. model of T_3 and let $C \leq M$ be a finite subgroup. Then there is $C \leq D \leq M$ such that*

1. D is generated by at most $15n^2$ elements where C is generated by at most n elements,
2. the lower central series of D coincides with its upper central series in reverse order. In particular, $D \leq_{\text{LCS}} M$.

Proof. If $C = 1$, take $D = 1$. Hence we may assume that $C \neq 1$. By Lemma 4.8, there is an extension $M \leq H_1$, $C \leq D_1 \leq H_1$ such that $D_1 \cap \gamma_2(H_1) = \gamma_2(D_1)$. The group D_1 is generated by C and at most $2n$ elements. Hence D_1 is generated by at most $3n$ elements. Next we apply Lemma 4.10 and get an extension $H_1 \leq H_2$ and $D_1 \leq D_2 \leq_{\text{LCS}} H_2$. D_2 is generated by D_1 and at most $3(3n)(3n-1)/2$ elements. Hence D_2 is generated by at most $3n + 9n(3n-1)/2$ elements.

Let $F_2 = F_{\mathcal{V}_3}(\{x, y\})$ and set

$$D_3 = D_2 *_{\mathcal{V}_3} F_2, \quad H = H_2 *_{\mathcal{V}_3} F_2.$$

Since $1 \neq D_2 \leq_{\text{LCS}} H_2$, Lemma 4.5 implies that $D_3 \leq H$. By Lemma 4.6, the lower central series of D_3 coincides with its upper central series in reverse order. D_3 is generated by at most

$$3n + \frac{9n(3n-1)}{2} + 2 \leq 15n^2$$

elements. Since M is an e.c. model and D_3 is finite, we can find $C \leq D \leq M$ such that $D \cong_C D_3$. Since D is a copy of D_3 , the lower central series of D coincides with its upper central series in reverse order. $\square$

5 Examples

In this section, we give examples showing that lower-central strictness and existential closedness cannot be omitted from the relevant results in Sections 3 and 4.

We start with an easy example.

Example 5.1. Let $G = F_{\mathcal{V}_3}(x, y, z) = B(3, 3)$. Then by Lemma 4.6 (or by a direct calculation), $\gamma_2(G) = Z_2(G)$ and $\gamma_3(G) = Z(G)$.

Let $a = [x, y] \in \gamma_2(G)$ and $A = \langle a, z \rangle \leq G$. Then, by Fact 2.27, $A \cong B(2, 3)$ and $a \notin \gamma_2(A) = \{[a, z]^k \mid k \in \mathbb{F}_3\}$. Let $F_2 = F_{\mathcal{V}_3}(v, w)$. Then

1. $A \not\leq_{\text{LCS}} G$,
2. The groups $A *_{\mathcal{V}_3} F_2$ and G do not have an amalgam over A in T_3 , because $[a, v, w] \neq 1$ in $A *_{\mathcal{V}_3} F_2$ but $[a, b, c] = [[x, y], b, c] = 1$ for any $b, c \in H \geq G$,
3. $A *_{\mathcal{V}_3} F_2 \rightarrow G *_{\mathcal{V}_3} F_2$ is not an embedding, because $[a, v, w] \mapsto 1$.

Remark 5.2. The example shows that lower-central strictness cannot be omitted from Lemma 4.5. In the proof of Theorem 3.3, Proposition A supplies a subgroup for which this injectivity holds, allowing the witness to failure of amalgamation to be transferred to a finite coproduct.

We next show that the uniform bounds in Theorem 3.3 and Proposition 4.13 fail without existential closedness.

Lemma 5.3. Let $F_\omega = F_{\mathcal{V}_3}(x_i \mid i \in \mathbb{N}) \in \mathcal{V}_3$. Let $n \geq 1$, $D \leq F_\omega$, and let $a_n := \prod_{i < n} [x_{2i}, x_{2i+1}] \in D \leq F_\omega$. If $a_n \in \gamma_2(D)$, then $d(D) \geq 2n$.

Proof. Put $V = \text{gr}_1(F_\omega) = \bigoplus_i \mathbb{F}_3 x_i$ and $U = \text{gr}_1^{F_\omega}(D) \leq V$. The image of $\gamma_2(D)$ in $\text{gr}_2(F_\omega) = \Lambda^2 V$ is contained in $\Lambda^2 U$, since $\gamma_2(D)$ is generated by commutators of elements of D . Hence $a_n \in \gamma_2(D)$ implies

$$\alpha_n := \text{gr}_2(a_n) = \sum_{i=0}^{n-1} x_{2i} \wedge x_{2i+1} \in \Lambda^2 U.$$

Let V^* be the vector space of all linear maps from V to $\mathbb{F}_3$. For $\lambda \in V^*$, define the contraction $\iota_\lambda : \Lambda^2 V \rightarrow V$ by

$$\iota_\lambda(u \wedge v) = \lambda(u)v - \lambda(v)u.$$

This is well-defined and linear, and $\iota_\lambda(\Lambda^2 U) \subseteq U$. Let $x_j^* \in V^*$ be the coordinate functional defined by $x_j^*(x_i) = \delta_{ij}$. For $0 \leq k < n$, we obtain

$$x_{2k+1} = \iota_{x_{2k}^*}(\alpha_n) \in U, \quad -x_{2k} = \iota_{x_{2k+1}^*}(\alpha_n) \in U.$$

Hence $\dim U \geq 2n$. Since the images in U of any generating set of D span U , we have $d(D) \geq \dim U \geq 2n$. $\square$

Lemma 5.4. *Let $1 \neq a \in G \in \mathcal{V}_3$ and $A = \langle a \rangle_G$. Let $F_3 = F_{\mathcal{V}_3}(s, x, y)$ and identify A with $\langle s \rangle$ via $a \mapsto s$. Then G and F_3 have an amalgam over A if and only if $a \notin \gamma_2(G)$.*

Proof. Suppose that $a \in \gamma_2(G)$. Then for any $H \in \mathcal{V}_3$ containing G , $[a, b, c] = 1$ for every $b, c \in H$. On the other hand, $[a, x, y] \neq 1$ in F_3 . This means G and F_3 do not have an amalgam H over A .

Conversely, suppose that $a \notin \gamma_2(G)$. There is a linear map

$$\lambda : \text{gr}_1(G) \rightarrow \mathbb{F}_3$$

sending $\text{gr}_1(a)$ to 1. Hence

$$r_G(g) := a^{\lambda(\text{gr}_1(g))}$$

defines a retraction $r_G : G \rightarrow A$. There is also a retraction $r_{F_3} : F_3 \rightarrow A$ ($x, y \mapsto 1$ and $s \mapsto a$). Hence $G \times F_3$ is an amalgam of G and F_3 over A through $G \ni g \mapsto (g, r_G(g)) \in G \times F_3$ and $F_3 \ni b \mapsto (r_{F_3}(b), b) \in G \times F_3$. $\square$

Let $F_\omega = F_{\mathcal{V}_3}(\{x_i \mid i \in \mathbb{N}\}) \in \mathcal{V}_3$, $n \geq 1$, $a_n = \prod_{i < n} [x_{2i}, x_{2i+1}]$ and let $A_n = \langle a_n \rangle \leq F_\omega$. (Clearly, $A_n \cong A_m$ for every $n, m > 0$ because A_n is non-trivial and $a_n^3 = 1$.) Let $F_3 = F_{\mathcal{V}_3}(s, y, z) \in \mathcal{V}_3$ and identify A_n with $\langle s \rangle \leq F_3$ via $a_n \mapsto s$. Then F_3 and F_ω do not have an amalgam over A_n by the previous lemma.

Proposition 5.5. *1. If $A_n \leq D \leq_{\text{LCS}} F_\omega$ then $d(D) \geq 2n$.*

2. For any $A_n \leq D \leq F_\omega$, if D and F_3 do not have an amalgam in $\mathcal{V}_3$ over A_n , then $d(D) \geq 2n$.

Proof. (1): Since $a_n \in \gamma_2(F_\omega)$, if $D \leq_{\text{LCS}} F_\omega$, then $a_n \in \gamma_2(D)$. Hence, by Lemma 5.3, $d(D) \geq 2n$.

(2): Since F_3 and D do not have an amalgam over A_n , by Lemma 5.4, $a_n \in \gamma_2(D)$. Hence, by Lemma 5.3, $d(D) \geq 2n$. $\square$

Since $d(A_n) = 1$ and $d(F_3) = 3$ for every $n \geq 1$, while the required numbers of generators are at least $2n$, neither the finite-envelope bound in Proposition 4.13 nor the bounded-obstruction conclusion of Theorem 3.3 extends to arbitrary models of T_3 .

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